Engineering ML project: crusher energy demand

CML3113 | Mining and Petroleum Engineering | Part A 30 marks + Part B 15 marks = 45 marks, scaled to 40 percent of the course | One 3-hour sitting, individual work, marked by computer

At a glance

The sitting: Friday 4 December 2026, 08:00 to 11:00 Kampala time (3 hours; proposed, lecturer to confirm). In Google Colab, with the same entry page, countdown and Submit test steps as Practical Test 1.

The rehearsal (unmarked): released Friday 27 November 2026 (after Class 4; proposed, lecturer to confirm). Same structure and the same kinds of task, on a practice data file. A solved walk-through is not given.

One notebook, one person. Everything is marked automatically, so nobody reads your notebook for marks. Part A is the coded notebook (30 marks). Part B is the decision memo (15 marks), written as a form inside the notebook.

It covers only what we taught in Weeks 1 to 11. There is nothing new to learn.

You get one of three data files (variants A, B and C), chosen for you at random when you start. Your numbers will differ from a classmate's.

1. The situation and the two questions

You work for the mine manager who pays for the crusher power. They give you one file, engineering_data.csv: 91 rows of invented (synthetic) readings from six pieces of equipment, E1 to E6. Each row has the load in percent, the ambient temperature in degrees Celsius and the crusher energy demand in kWh/t (the target variable). Some cells are missing and some rows are repeated or impossible, as in Practical Test 1. You answer two questions.

	What you must answer
Question 1 **How big?**	How many kWh/t of crusher energy demand will a crusher show at a given load and ambient temperature, for a crusher that the model has never met? Compare a baseline model, a multiple linear regression model and a decision tree of depth 4, on the same rows, with MAE and RMSE in kWh/t.
Question 2 **Warn or not?**	A row is a warning (1, the positive class) when its crusher energy demand is at or above 13.5 kWh/t, and no warning (0) below it. Which rows should raise a warning? Give each row a probability with logistic regression, choose a decision threshold, and report the confusion matrix counts, precision, recall, F1, the cost and the ROC AUC. A false negative (a missed warning) costs 5 and a false positive (a false alarm) costs 1 (invented teaching costs).
The client's limits	A typical prediction error of at most 1 kWh/t (MAE) and a prediction time of at most 50 ms. Your notebook turns this into a rule that picks one model. Both numbers are invented for teaching.

2. The data and the test set

- The supplied setup cell in your notebook creates your engineering_data.csv, so you need no upload.
- **Develop on E1 to E5.** Train on E1 to E4 (the training set), validate on E5 (the validation set), and use group cross-validation. The notebook gives you the E1 to E5 rows as development_records.
- **E6 is the test set.** It is a different piece of equipment that no part of your work may use until the very end. It gives one honest estimate for the model you chose.

How the test set is opened once

The supplied cell "Open the test set" is in your notebook. Do not edit it. It defines open_test_set(model_name, threshold). Near the end you call it ONE time: test_result = open_test_set(chosen_model, chosen_threshold), after you have chosen the model and the threshold on the validation set. It trains your own functions on all of E1 to E5, evaluates on E6 and prints the result.

The notebook counts how often you call it. A second call loses the "opened once" mark, even with the same choice. A second call with a different model or threshold also prints a warning that is saved in your notebook and loses the test set marks.

Never use E6 to choose a model, a threshold or a hyperparameter. Once you have looked at it and then changed your mind, it can no longer tell you how the model behaves on new equipment.

3. How the sitting works

1. **Before you start.** Bring a charged laptop and your own Google account. Open the entry link or scan the QR code your lecturer shows. Check the course and the title "Engineering ML project". Do not start until your lecturer says so.
2. **Start.** Enter your full name, your Google email and your Student Registration Number, then press Start test. Your three hours begin at that moment. Opening the page alone does not start the clock. Use the same Google account in Colab and in the form.
3. **Your notebook.** Press Open my test in Colab. It is your own private notebook; no copy is needed. Run the first code cell (your details and the countdown), then the supplied setup cell. Work from the top. Save often with File, then Save.
4. **The deadline.** Your deadline is the earlier of three hours after you started and the class closing time. The clock cannot be paused. Refreshing, a new device or an internet interruption does not restart it. A student who starts late has only the time left until the class closing time.
5. **The end.** At your deadline the Submit test button switches itself off, the lecturer's server refuses new snapshots, and a form response after the deadline is kept for review and is not marked. Colab may still let you type; nothing you type after the deadline is marked.
6. **If something fails.** Put your hand up and tell your lecturer at once. Do not make a second identity and do not use another student's file. If Colab disconnects, reconnect and run the countdown cell and the setup cell again; your saved notebook is still in Drive and the timer continues.

Suggested time plan	Time
Setup: sign in, open your notebook, run the supplied cells	10 min
Tasks 1 to 3: cleaning, the split, the three models	45 min
Tasks 4 to 6: warnings, cross-validation, ablation	45 min
Tasks 7 and 8: choose a model, open the test set once	15 min
Task 9: the Part B memo form	25 min
Check, restart and run all, submit	25 min
Spare time (breaks, reconnecting)	15 min
Total	180 min

4. What you hand in, exactly

Item	Detail
One notebook	Your issued notebook, with Tasks 1 to 9 done (the Part B memo form is Task 9). Restarted, run from the top, then saved.
How	The last cell of the notebook. Run it, click Submit test, open the form link, check your name and registration number, and press Submit. Wait for the message that your response was recorded. Allow at least 20 minutes for this step.
Nothing else	No ZIP, CSV, PDF, Word file, screenshot or separate memo. One submission per person. A new submission before your deadline must be the complete notebook; the latest one on time is marked.

5. Allowed tools and the AI rule

These are the same rules as Practical Test 1.

- **Permitted:** Google Colab, your notebook and its CSV, Python with pandas, NumPy, matplotlib and scikit-learn as used in class, the submission form, and your lecture slides, your lab notebooks from Classes 1 to 4, the revision guide, the cheatsheet and your own course notes.
- **No AI or outside help:** do not use Gemini, ChatGPT, Copilot or any other AI tool to generate, complete, rewrite, explain or debug your answers. Do not get answers from another person, collaborate, message or copy someone else's work.
- **Understanding check:** you may be asked to explain your submitted code or make a small change to it, without AI or another person's help.
- **Declaration:** by submitting, you confirm that you completed this project yourself without AI assistance or another person's help, using only the permitted resources, and that you can explain and change the code you submitted.

The lecturer's computer compares notebooks for copying and lists unusual cases for a person to look at. A flag never changes a mark by itself.

6. Rules that make your notebook run every time

1. **Restart and run all.** Choose Runtime, then Restart session and run all, before you submit. Every cell must run from the top without an error, in under three minutes in total and under 30 seconds per cell.
2. **Keep the supplied cells and names.** Do not edit the supplied cells (the ones that say Do not edit this cell). Keep each function name, argument order, returned keys and variable name exactly as printed.
3. **Use only the arguments.** A function uses the lists it is given, never a global table. It is marked on other data you cannot see.
4. **Train on the training set only.** The imputation, the StandardScaler and the baseline model's mean are learned from the training rows. Using validation or test rows in them is data leakage and loses marks.
5. **Seeds.** Keep random_state=17 for the decision tree. Do not add random choices without a seed.
6. **Calculate, do not type.** Your results come from your code. A number typed in by hand earns nothing in Part A.
7. **No extras.** No pip installs, no downloads, no extra files.

7. How Part A is marked (30 marks)

The lecturer's computer runs your notebook from the top in a locked box. The self-check cells in your notebook are not marked: they use tiny examples with known answers and show PASS or TODO. The hidden checks run your functions on other data generated for your course and compare the numbers; they also read your named variables for your own data file. You cannot see them, so test your functions with your own small examples too. Each task earns the share of its checks that pass, so one wrong task does not cost the others. Six of the 30 marks are for how you worked (items A12 to A15).

Part A item	What is checked	Taught in	Marks
A1. Clean the rows	Your function on messy data it has not seen: first copy kept, missing target and loads outside 0 to 100 dropped, ambient_c left as a missing value, your input unchanged.	Weeks 3 to 7	2
A2. Split by equipment group	Your function on other rows, also in a different order: E1 to E4 training set, E5 validation set, E6 test set, no use of the target.	Week 10	2
A3. Baseline model, linear regression and decision tree	Baseline model, linear regression and decision tree predictions on other data with missing values, fitted on training rows only, with the median imputation learned from training rows.	Weeks 8 and 9	3
A4. MAE and RMSE	MAE and RMSE on other numbers, including a case with no error.	Week 8	2
A5. Warning probabilities (logistic regression)	Warning probabilities (logistic regression) on other data, for the limit you are given and for another limit.	Week 9	2
A6. Confusion matrix, precision, recall, F1, cost and AUC	Confusion matrix counts, precision, recall, F1, cost and AUC at four decision thresholds, including one where nothing is flagged.	Week 9	3

Part A item	What is checked	Taught in	Marks
A7. Group cross-validation	Five group folds, their mean and their spread, on other data.	Week 10	3
A8. Feature ablation	Base error, error with the extra feature, and the improvement, on other data.	Week 11	2
A9. Choose a model by the rule	The selection rule, including ties and the case where nobody qualifies.	Week 11	1
A10. Your results on the supplied file	The numbers in your notebook for your data file (validation errors, cross-validation, AUC, ablation, your chosen model) against the correct values. Numbers typed in by hand earn nothing.	Weeks 8 to 11	2
A11. The test set result	The test set opened with your chosen model and threshold; the supplied cells unedited; the test result is the correct one for your choice.	Week 11	2
A12. Test set opened once, after the choice	The test set opened exactly once, after you chose the model and the threshold, with that choice; no test-set row among your development rows.	Weeks 10 and 11	2
A13. No data leakage: preprocessing fitted on training data only	Four probes: the imputation, the baseline model, the StandardScaler and the group cross-validation use only the rows they are given for training (no data leakage).	Weeks 8 to 10	2
A14. Reproducible with the seed	Calling your regression function twice gives the same answer; the decision tree keeps random_state=17; no unseeded randomness; your saved predictions equal a fresh run.	Weeks 8 to 11	1
A15. Charts and pipeline requirements	Chart 1 and Chart 2 each have a title, named axes and the unit, and plot the right data; every required variable exists.	Weeks 3 to 8	1
Total			30

8. Part B: the decision memo form (15 marks)

The memo is for the mine manager who pays for the crusher power, who is not a data scientist. You write it as a form: a Python dictionary called MEMO in Task 9. Every field is checked automatically, so type your numbers carefully (two decimals are enough, in kWh/t). The lists of limits and next steps are printed in your notebook; each student sees their own lettering.

Field	What you enter and how it is checked	Marks
recommended_model, recommended_threshold	A model you trained ("baseline", "linear" or "tree") and a decision threshold (0.3, 0.5 or 0.7). They must be the choice your own rule and cost table made.	3
validation_mae	The validation MAE of the three models. Compared with the values your own notebook computed.	3
test_mae, test_recall	The MAE and the warning recall on the test set from your one opening. Compared with the value the marking computer calculates from the same predictions.	2
cv_mean_mae	The mean MAE of your group cross-validation. Compared with your own cv_summary.	1
limits	Three different letters: the statements that are true limits of this study. Ten statements are listed and six are false. One mark for each true one.	3
next_step	One letter: the best next step, from five.	1
answer_choice, answer_test_set	Two short written answers of 8 to 80 words each: why you recommend your model (use a validation number and say what you compared it with) and why the test set is opened only once. One mark each. Several correct wordings are accepted; write sentences in your own words.	2

Field	What you enter and how it is checked	Marks
Total		15

9. Rehearsal, help and the day

- Rehearsal: Friday 27 November 2026 (after Class 4; proposed, lecturer to confirm). Do it with a timer. It is unmarked. Ask about anything unclear before the sitting.
- Sitting: Friday 4 December 2026, 08:00 to 11:00 Kampala time (3 hours; proposed, lecturer to confirm).
- Your lecturer answers questions about task wording and about ideas from Classes 1 to 4. Your lecturer does not write or debug your answers.
- Practise the whole path in the rehearsal: restart and run all, fill in the MEMO form until the memo self-check shows 10 of 10, and read the self-check output.

Before you submit

☐	Identity cell correct (name, email, registration number). COHORT, TEST_ID and VARIANT unchanged.
☐	Restart session and run all: no error, every cell ran.
☐	The test set was opened one time, after the choice. Chart 1 and Chart 2 have a title, named axes and the unit.
☐	The MEMO form is complete and the memo self-check passes.
☐	File saved, Submit test clicked, form submitted, confirmation seen before your deadline.